

Math 340: Elementary Matrix and Linear Algebra

MIDTERM TWO

Wednesday, November 15th, 2023, 7:30pm-9:00pm

Please circle your Instructor and your Discussion Session (date and start time given) in the table below.

Dr. Kaitlyn Phillipson	Dr. Shengwen Gan	Dr. Joshua Mundinger	Dr. Feng Zhu
Joey (Yu), T 3:30 PM	Zaidan, M 12:05 PM	Katerina, M 7:45 AM	Anna, T 1:20 PM
Joey (Yu), T 4:35 PM	Zaidan, W 12:05 PM	Katerina, M 8:50 AM	Anna, T 2:25 PM
Joey (Yu), R 3:30 PM	Daniel, M 12:05 PM	Ruitao, W 7:45 AM	Anna, R 2:25 PM
Yaoxian, R 4:35 PM	Daniel, W 1:20 PM	Ruitao, W 8:50 AM	
Hongyu, R 9:55 AM	Dewei, M 3:30 PM	Jiaqi, M 11:00 AM	
Hongyu, T 9:55 AM	Dewei, W 3:30 PM	Jiaqi, W 12:05 PM	
Hongyu, R 8:50 AM	Dewei, M 2:25 PM	Ellie, W 1:20 PM	
	Joel, W 4:35 PM	Ellie, M 1:20 PM	

Name: _____ Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of nine problems for a total of 50 points; the exam is 6 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices with standard matrix addition and scalar multiplication, P_d is the vector space of polynomials (in x) of degree at most d with standard polynomial addition and scalar multiplication, $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (5 points, 1 point each) Are each of the following true or false statements? Fill in the bubble next to the correct answer. Justification is not necessary.

a.) If U is a subspace, and $a\mathbf{u}$ is in U for some nonzero scalar a , then $\mathbf{u}$ is also in U .

- ☐ True
☐ False

b.) If A is similar to B , then A and B have the same determinant.

- ☐ True
☐ False

c.) It is possible to create a one-to-one linear transformation $T : M_{22} \rightarrow P_2$.

- ☐ True
☐ False

d.) A set of six linearly independent matrices in M_{32} will also span M_{32} .

- ☐ True
☐ False

e.) If $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly independent set, so is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2\}$.

- ☐ True
☐ False

2. (5 points) Consider $\mathbb{R}^3$ and the following vector addition and scalar multiplication:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \oplus \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 + 3 \\ x_3 + y_3 \end{bmatrix}, \quad c \odot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 + 3c - 3 \\ cx_3 \end{bmatrix}$$

a.) (3 points) Find the zero vector $\mathbf{0}$ such that $\mathbf{v} \oplus \mathbf{0} = \mathbf{v} = \mathbf{0} \oplus \mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^3$. Fully justify your answer.

b.) (2 points) For $\mathbf{v} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, find the additive inverse of $\mathbf{v}$ (i.e. find $\mathbf{u}$ such that $\mathbf{u} \oplus \mathbf{v} = \mathbf{0} = \mathbf{v} \oplus \mathbf{u}$). Fully justify your answer.

3. (6 points total) For $A = \begin{bmatrix} 4 & -2 & 2 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix}$:

a.) (4 points) Find a basis for the eigenspace associated to the eigenvalue $\lambda = 2$ (in other words, find a set of basic eigenvectors associated to $\lambda = 2$). Show all work.

b.) (2 points) Given that $\lambda = 3$ is the only other eigenvalue of A , and a basis for the eigenspace of $\lambda = 3$ is $\left\{ \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \right\}$, answer the following: Is A diagonalizable? Fully justify your answer.

4. (5 points) Consider the subspace W of $\mathbb{R}^3$, where

$$W = \left\{ \begin{bmatrix} a+2b-2c \\ -3a+b-8c \\ a+4b-6c \end{bmatrix} \middle| a, b, c \text{ are in } \mathbb{R} \right\}.$$

Find a basis for W . (Hint: Start by finding a spanning set for W). Show all work.

5. (5 points) Suppose $T : P_1 \rightarrow M_{22}$ is a linear transformation with

$$T(x-1) = \begin{bmatrix} 2 & -4 \\ 0 & 1 \end{bmatrix}, T(x+2) = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}.$$

Find $T(7x+5)$. Show all work.

6. (5 points) Consider the vector space M_{22} with standard matrix addition and scalar multiplication.

Consider the subset W of M_{22} of matrices of the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b = 0$, $c - d = 0$, i.e.,

$$W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a + b = 0, c - d = 0 \right\}.$$

Is W a subspace of M_{22} ? Completely justify your answer.

7. (4 points) For which c will the set $\left\{ \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ c \end{bmatrix} \right\}$ NOT form a basis for $\mathbb{R}^3$? Completely justify your answer.

8. (9 points) For this problem, use A and the reduced row echelon form of A , $RREF(A)$, to answer the following questions.

$$A = \begin{bmatrix} 1 & 1 & -1 & 1 & 4 \\ 0 & 1 & -3 & -2 & -8 \\ 1 & 0 & 2 & 3 & 12 \\ 2 & 1 & 1 & -1 & -4 \end{bmatrix} \quad RREF(A) = \begin{bmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a.) (1 point) What is the rank of A ? Explain your answer in one sentence.
- b.) (2 points) Find a basis for the nullspace of A . What is the dimension of the nullspace of A ?
- c.) (2 points) Find a basis for the span of the column vectors of A (in other words, a basis for the column space of A). Explain your answer in one sentence.
- d.) (2 points) Consider the linear transformation $T : \mathbb{R}^5 \rightarrow \mathbb{R}^4$ by $T(\mathbf{x}) = A\mathbf{x}$ (yes, this is the same A as above). Is T *one-to-one* (also called *injective*)? Explain your answer in one sentence.
- e.) (2 points) Consider the linear transformation $T : \mathbb{R}^5 \rightarrow \mathbb{R}^4$ by $T(\mathbf{x}) = A\mathbf{x}$ (again, this is the same A as above). Is T *onto* (also called *surjective*)? Explain your answer in one sentence.

9. (6 points total) Each of the following statements is **false**. Show each statement is false by providing explicit counterexamples. Fully justify your answers.

a. (2 points) The function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $F \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 x_2 \\ 0 \end{bmatrix}$ is a linear transformation.

b. (2 points) Let V be $\mathbb{R}^2$ with the operation $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \oplus \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}$. $\oplus$ satisfies the vector space property $\mathbf{v} \oplus \mathbf{u} = \mathbf{u} \oplus \mathbf{v}$.

c. (2 points) The set of vectors $W = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \mid a + b \geq 0 \right\}$ is a subspace of $\mathbb{R}^2$.

Scratch paper page!!

Back of scratch paper page!